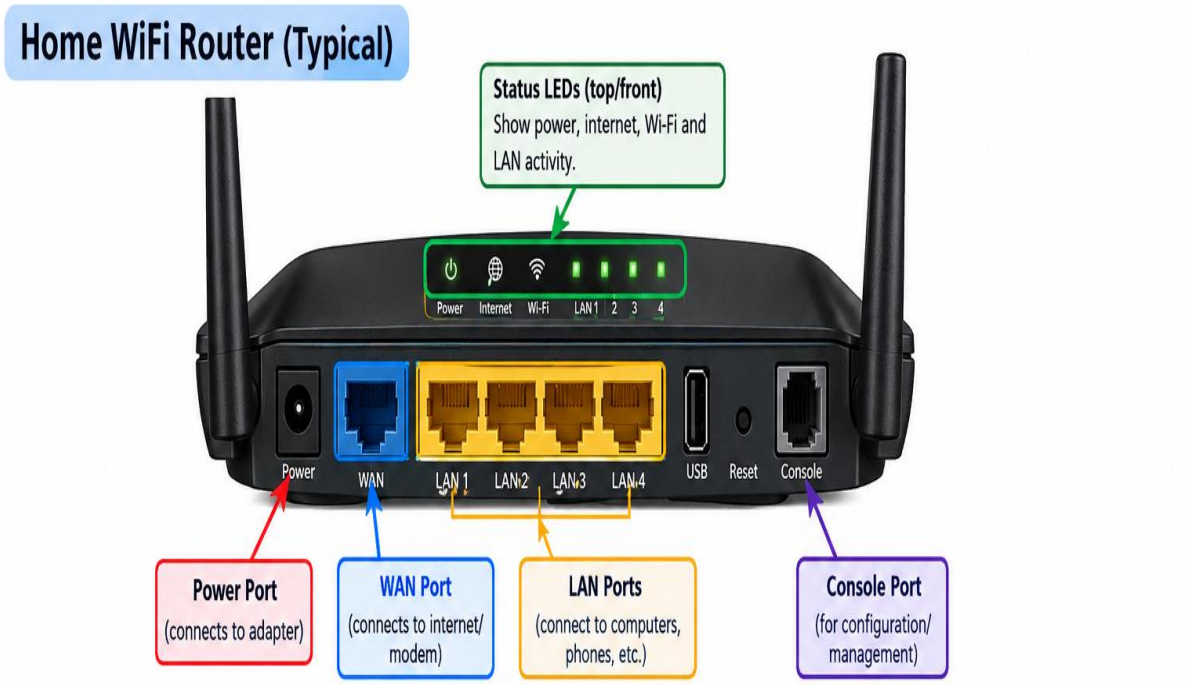
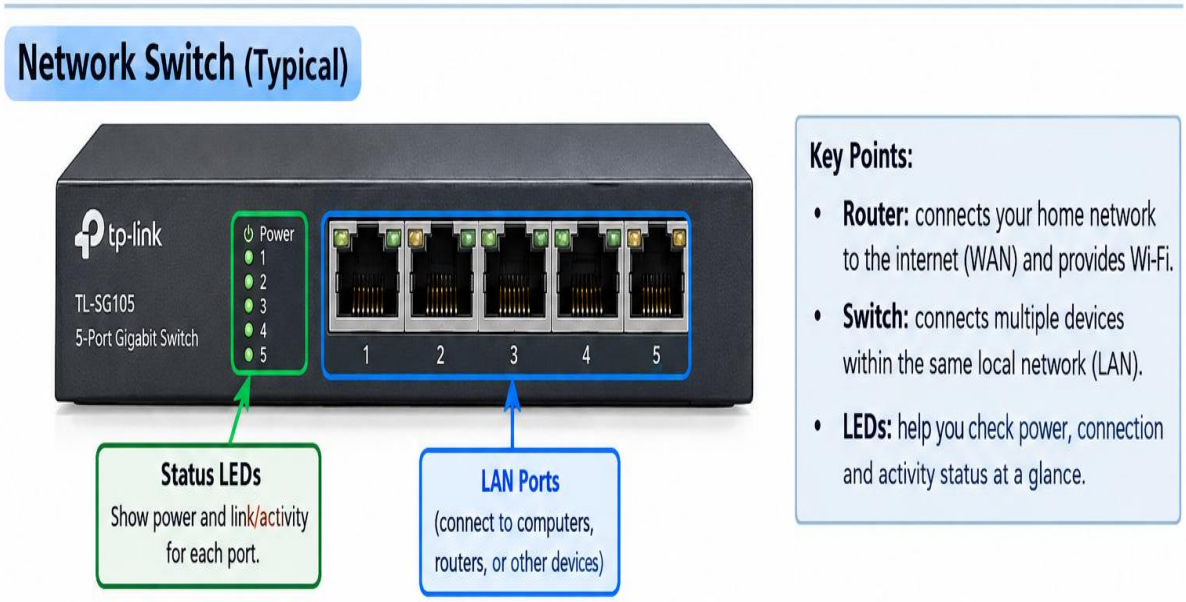
1.Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

Ans.





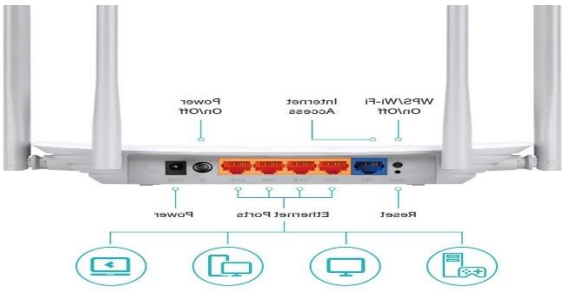
2.List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Ans.

No.	Router	Switch
1. Main Function	A router connects different networks, such as a home network to the Internet.	A switch connects multiple devices within the same local network (LAN).
2. Address Used	A router mainly uses IP addresses to forward data packets to the correct network.	A switch mainly uses MAC addresses to forward data frames to the correct device.
3. Network Traffic	A router directs traffic between different networks and selects a suitable path for data.	A switch sends data to the required device within the same network, reducing unnecessary traffic.

3.Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

Ans. A. Home Wi-Fi Router



1. **Power Port**: Usually located on the back of the router. It connects the power adapter to supply electricity to the device.
2. **Ethernet Ports (WAN and LAN):** The WAN port connects the router to the modem or Internet connection. The LAN ports connect computers and other wired devices to the router.
3. **LED Indicators:** Usually located on the front or top of the router. They indicate power status, Internet connection, Wi-Fi activity, and LAN port activity.

B. Network Switch



1. **Power Port**: Usually located on the back or side of the switch. It connects the power adapter to provide electricity.
2. **Ethernet Ports**: Located on the front of the switch. These ports connect computers, printers, routers, and other network devices using Ethernet cables.
3. **LED Indicators**: Located near the Ethernet ports. They show whether the switch has power and whether each port has an active network connection or data activity.

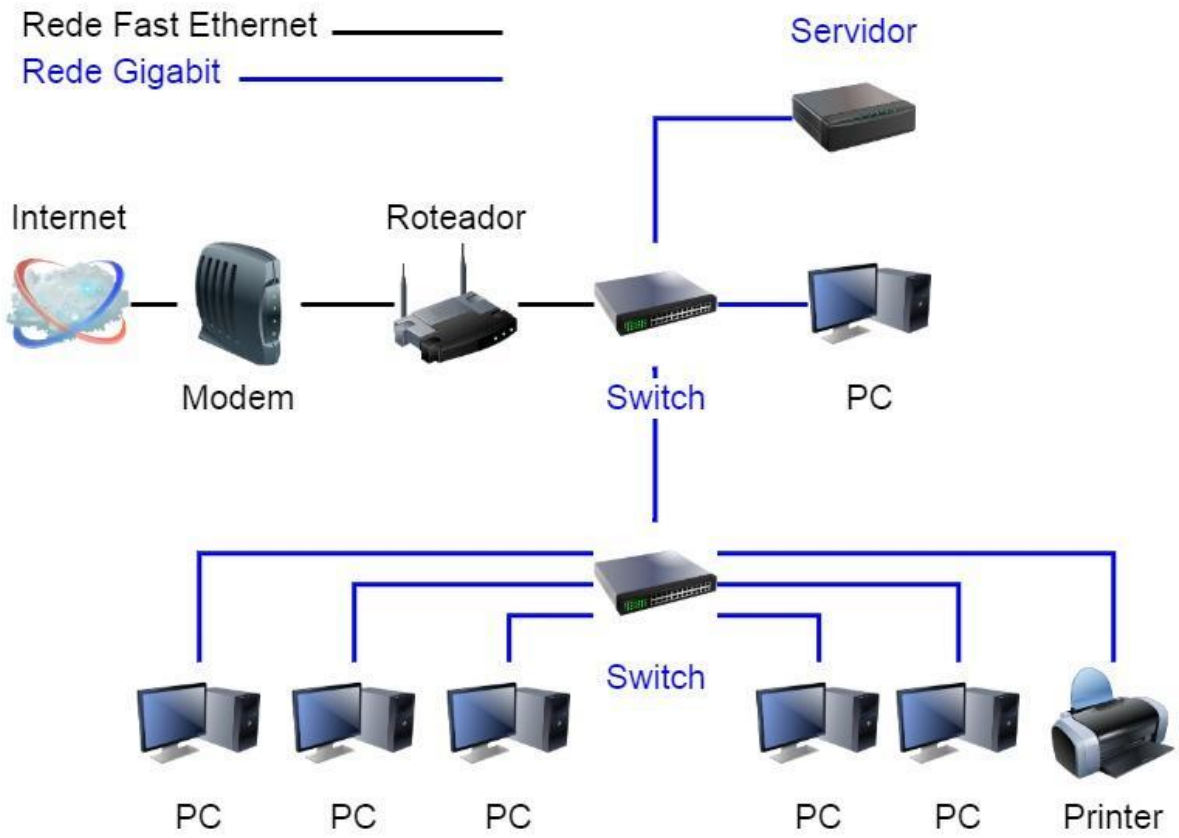
4.Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Ans.

Scenario:

I am setting up a small gaming café with 8 gaming PCs for playing multiplayer games and watching live IPL matches. First, I would connect the ISP (Internet Service Provider) line to the modem and then connect the modem to the WAN port of the router using an Ethernet cable. The router would provide Internet access and manage communication between the café network and the Internet.

Next, I would connect one LAN port of the router to an Ethernet port on a network switch using a network cable. Then, I would connect all 8 gaming PCs to the switch using separate Ethernet cables.



Justification:

1. **Router**: It connects the café network to the Internet and manages traffic between the local network and external networks**.**

2. **Network Switch**: It provides enough Ethernet ports to connect all 8 gaming PCs to the same local network.
3. **Ethernet Cables:** Wired connections provide stable connectivity, low latency, and less interference, which are important for multiplayer gaming and smooth IPL streaming.